

enhance solar cell efficiency. High losses of photons due to both incomplete absorptions of the solar spectrum and thermalisation are the main drawbacks that have been addressed to overcome limitations in photovoltaics efficiency.

Surface plasmon resonance (SPR) arises in metal nanoparticles (NP) as a collective oscillation of the surface electrons when subjected to the electric field associated with the incident light. SPR is the resonant oscillation of conduction electrons at the interface between negative and positive permittivity material stimulated by incident light. Permittivity is a measure of the electric polarisability of a dielectric. Also, NPs should be smaller than the wavelength of the incident light, enabling them to polarise in order to achieve SPR.

The scattering and absorption of light depend on the polarisation of the NPs, which is a function of the size, refractive index, shape, density, and the surrounding medium. These parameters should be tuned in order to maximise the NPs' light scattering relative to absorption. Scattering lower than fifty per cent may lead to heat dissipation and absorption suppression. This promotes the undesired reduction of photocurrent for solar

cells having a high response in the blue region.

NPs used as a front electrode lead to an enhancement of the intrinsic layer absorption above 59 per cent in another light spectrum like the infrared region. Silicon does not absorb light very well. For this reason, more light needs to be scattered across the surface in order to increase the absorption.

It has been found that metal nanoparticles like gold NPs help scatter the incoming light across the surface of the silicon substrate. A common design is to deposit gold NPs on the top surface of the silicon wafer, as shown in Fig. 5.

When light hits these gold NPs at their surface plasmon resonance, it gets scattered in many different directions. This allows light to travel along the silicon wafer and bounce between the substrate and NPs, enabling the solar cell to absorb more light. The concentrated near-field intensity induced by localised surface plasmon of the NPs promotes the optical absorption of the semiconductor.

The scattering cross-section of 15nm gold NPs embedded in silicon is many times (roughly thirty times) larger than its geometric cross-section. Therefore small surface coverage with these particles can highly scatter the incident light.

A simple and compatible process like thermal evaporation assisted by an electron beam to deposit gold NPs directly on the silicon wafer can be used. The layer thickness is controlled after opening the shutter of the system by the deposition time and a cali-

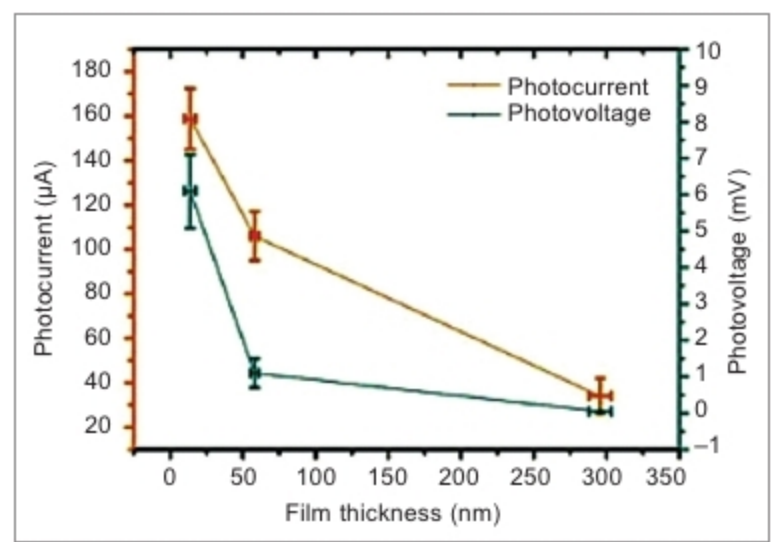


Fig. 7: Effect of gold film thickness on photoresponse

brated quartz crystal detector.

There is a decrease in transmittance as the gold NPs' thickness increases, while the diffused light is enhanced for higher thickness. At 15nm thickness of gold NPs, the transmittance is about eighty to ninety per cent, and reflectance is about fifteen per cent.

Photocurrent under partial illumination

When one half of the gold surface is selectively illuminated and the other half is blocked from light, photocurrent flows. When the dark and bright zones are swapped in reverse mode, a photocurrent of equal magnitude but of an opposite polarity flows.

There is a directional electron flow on the surface that always flows from the bright to the dark zone. This indicates a change in surface potential induced by illumination.

On the other hand, when the entire surface is uniformly illuminated, the photocurrent and photovoltage are significantly low.

Effect of gold film thickness

There is a 33 per cent reduction in photocurrent upon increasing the gold film thickness from 15nm to 60nm. And there is more than a four-fold drop in photoresponse when the thickness is further increased to 300nm under same lighting condition using white light of $\sim 18\text{mW/sq cm}$. The photovoltage also gets reduced, as shown in the curve (Fig. 7).

This is suggestive of the role of the underlying silicon substrate in driving photoresponse and the need for the gold film to be thin enough for the photons to reach the silicon interface. **EFY**

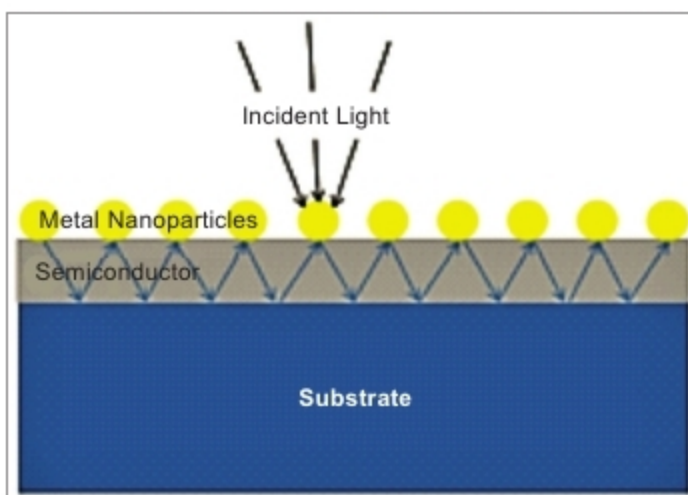


Fig. 5: Scattering of light by gold nanoparticles

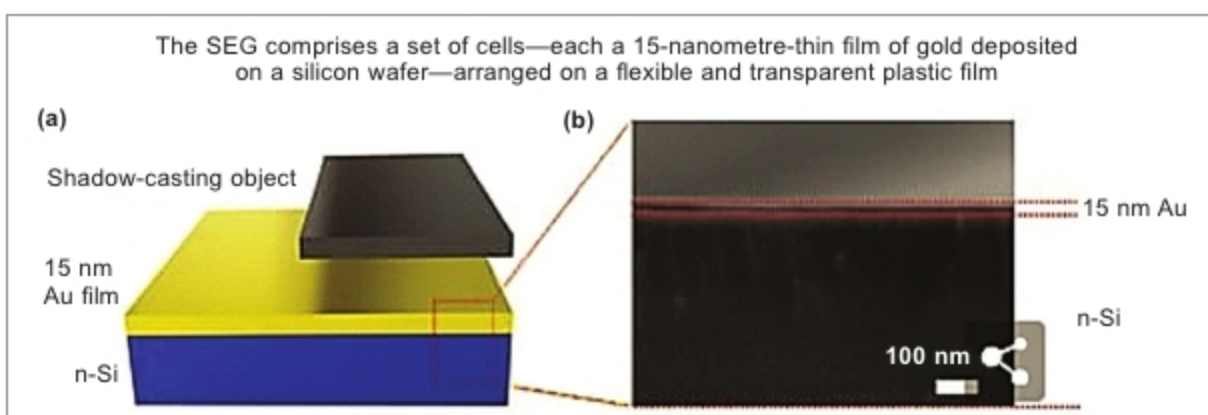


Fig. 6: Schematic diagram of a SEG with a thin film of gold NPs